Section: B1 + B2 **Duration: 1 hour**

Use of STL/library functions is allowed for both problems.

Problem 1

Once there was an Emperor named Akbar. He had a son named Jahangir. For an unforgivable reason, the king wanted him to leave the kingdom. Since he loved his son, he decided his son would be banished to a new place. The prince became sad, but he followed his father's will. On the way, he found that the place was a combination of land and water. Since he didn't know how to swim, he was only able to move on land. He didn't know how many places might be his destination. So, he asked for your help.

For simplicity, you can consider the place as a rectangular grid consisting of some cells. A cell can be a land or can contain water. Each time the prince can move to a new cell from his current position if they share a side.

Now write a program to find the number of cells (unit land) he could reach including the cell he was initially in.

Input

Input starts with a line containing two positive integers W and H; W and H are the numbers of cells in the x and y directions, respectively. W and H will not be more than 20.

There will be H more lines in the data set, each of which includes W characters. Each character represents the status of a cell as follows.

- . - land.
- # - water.
- @ - initial position of the prince (appears exactly once in a dataset).

Output

Print the case number and the number of cells he can reach from the initial position (including self).

Sample Input	Sample Output	Explanation
7 7 ..#.#.. ..#.#.. ###.### ...@... ###.### ..#.#.. ..#.#..	13	..#.#.. ..#.#.. ###.### ...@... ###.### ..#.#.. ..#.#..
6 9 ....#. .....# .... #@...# .#..#.	45	

Problem 2

You have to complete n courses. There are m requirements of the form "course a has to be completed before course b". Your task is to find an order in which you can complete the courses.

Input

The first input line has two integers n and m: the number of courses and requirements. The courses are numbered from 1 to n. (n, m <= 10^5)
After this, there are m lines describing the requirements. Each line has two integers a and b: course a has to be completed before course b.

Output

Print an order in which you can complete the courses. You can print any valid order that includes all the courses. If there are no solutions, print -1.

Sample Input	Sample Output	Explanation
5 3 1 2 3 1 4 5	3 4 1 5 2	
6 6 1 2 2 3 4 3 4 5 5 6 6 4	-1	The dependency among courses 4, 5 and 6 cannot be resolved.